Maximilian Puelma Touzel, PhD

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Staff Research Scientist,
Mila - Quebec AI Institute

Education

2011–2015 **PhD – Physics**, *University of Goettingen*, Germany, International Max Planck Research School in the Physics of Biological and Complex Systems
Dissertation: Cellular dynamics and stable chaos in balanced networks (Advisor: Fred Wolf)

2008–2009 **Master of Science – Physics**, *University of Toronto*, Canada

2001–2006 **Honours Bachelor of Science – Mathematics & Physics (Double Specialist)**, *University of Toronto*, Canada

Research Interests

Theoretical biophysics: statistical physics of adaptive biological systems; inference of stochastic population dynamics from high-throughput data (immune repertoires); dynamics and computation in neural circuits.
Theory of AI-agent collectives: informational active matter and information thermodynamics of agent populations; self-organized roles and incentives; collusion, monitoring and certification of interacting AI agents.
Collective & cognitive dynamics: decision-making under uncertainty; learning in multi-agent systems; collective behaviour across scales, from cells to brains to societies.
Methods: statistical inference, stochastic processes, dynamical systems & Lyapunov analysis, reinforcement learning, deep learning, NLP.
Applications: immunology & vaccination, neuroscience, and socio-technical systems (polarization, misinformation, AI safety).

Research Positions

2025–present **Staff Research Scientist, Applied Machine Learning Research Team**, *Mila*, Montréal, Canada
- Principal Investigator, Future of Life Institute program on AI-driven power concentration (2025–2027; no-cost extension funding students beyond Sept. 2027)
- Co-supervision of graduate students (with R. Rabbany)

2024–2025 **Complex Data Lab Research Scientist**, *Mila, McGill, Université de Montréal*, Montréal, Canada
- Project Lead: information integrity experiments with LLM-based simulations of social systems
- RAG-based LLM fact-checking and public-facing webtool (Veracity)
- Statistical inference of polarization and ideology from social-media network and text data

2022–2024 **Research Manager, Canadian Excellence Research Chair in Autonomous AI**, *Mila, Université de Montréal*, Montréal, Canada
- Scientific supervisor, advisor and liaison for the CERC group
- Team Lead, Agent Abstraction project (multi-agent reinforcement learning)

2020–2022 **Research Associate, Canadian Excellence Research Chair in Autonomous AI**, *Mila, Université de Montréal*, Montréal, Canada
- Scalable continual reinforcement learning (NeurIPS 2022)

2018–2020 **IVADO Post-Doctoral Fellow**, *Mila, Université de Montréal*, Montréal, Canada, Advisors: Yoshua Bengio (DIRO) & Guillaume Lajoie (DMS)

- Dynamical-systems theory of recurrent neural networks: transient dynamics and Lyapunov spectra (NeurIPS 2019; Front. Appl. Math. Stat. 2022)
- Normative models of urgency in human and primate decision-making, with P. Cisek (PLoS Comput. Biol. 2022)
- NeuroAI community organizer (reading group, NeurIPS workshop)

2015–2018 **European Research Council-funded Post-Doctoral Fellow**, *Laboratoire de physique théorique, École normale supérieure*, Paris, France, Advisors: Aleksandra Walczak & Thierry Mora

- Statistical inference of stochastic clonal dynamics in immune repertoires from high-throughput sequencing (PLoS Comput. Biol. 2020)
- Tracking vaccine-responding T-cell clones in identical twins (PNAS 2018); public B-cell clones after vaccination in fish (Front. Immunol. 2018)
- Probabilistic models of V(D)J recombination and selection; open-source expansion-detection tool NoisET (J. Phys. Chem. A 2022)

2010–2015 **IMPRS Excellence Award Doctoral Researcher**, *Theoretical Neurophysics Group, Max Planck Institute for Dynamics and Self-Organization*, Goettingen, Germany, Advisor: Fred Wolf

- Statistical physics of spiking networks: stable chaos and phase-space partitioning (Phys. Rev. E 2019)
- Dynamic response theory of neurons with complex intrinsic dynamics (PLoS Comput. Biol. 2015)

2009–2010 **Master's Researcher (experimental biophysics)**, *Systems Biophysics Lab, Department of Physics, University of Toronto*, Toronto, Canada, Advisor: William Ryu

- Design of thermotaxis experiments and confocal imaging of *C. elegans*

2004–2005 **Undergraduate Researcher**, *Centre for Quantum Information and Quantum Control, Department of Physics, University of Toronto*, Toronto, Canada, Advisor: Aephraim Steinberg

- Optimal measurement theory in quantum state discrimination (Phys. Rev. A 2007)

Awards & Grants

2025 **Mitigate AI-driven Power Concentration**, *Future of Life Institute*, main PI, 2 years
$620,000

2024 **Exploratory Grant**, *IVADO*, co-PI, 1 year
$100,000

2024 **Team Grant**, *Foresight Institute*, co-PI, 1 year
$100,000

2021 **Team Grant**, *Fonds de recherche du Québec – Nature et technologies (FRQNT)*, co-applicant, 5/2021–4/2024
$150,000. *Études de la prise de décision dynamique en milieu naturel et artificiel: Une approche interdisciplinaire et intégrative* (PI: P. Cisek, with K. Jerbi, A. Green, G. Lajoie, I. Rish, M. Puelma Touzel)

2019 **Conference Poster Award**, *Montreal AI & Neuroscience Conference*

2018 **Post-doctoral Fellowship Award**, *IVADO*, 2 years

2014 **Summer School Start-up Program grant award**, *University of Goettingen*, Lead

2012 **Conference Poster Award**, *ACCN*

2011 **PhD Excellence Fellowship Award**, *IMPRS*, 3 years

Publications

25 peer-reviewed works; 8 first-author journal and main-conference papers. Numbering is per section.

Preprints & Work Under Review

[3] Andreea Musulan, Zachary Yang, Ardysatrio Haroen, Victor Livernoche, Taylor Lynn Curtis, Maximilian Puelma Touzel, Anne Imouza, Roxanne Corriveau, Kellin Pelrine, Eric Boxer, Jean-François Godbout, and Reihaneh Rabbany. *Malicious Coordinated Activity on X: A Case Study of the 2025 Canadian Federal Election*. Under review. 2026.

[2] Maximilian Puelma Touzel. *Measurement-Induced Role Differentiation in Agent Populations*. Updated version, November 2026. 2026. arXiv: 2609.05442.

[1] Maximilian Puelma Touzel and Miloš Broćić. *A Simulation Framework for Systemic Impact Evaluation*. Under review. 2025.

Biological Physics & Theoretical Neuroscience

[10] Meriem Bensouda Koraichi, Maximilian Puelma Touzel, Andrea Mazzolini, Thierry Mora, and Aleksandra M Walczak. "NoisET: Noise Learning and Expansion Detection of T-Cell Receptors". In: *The Journal of Physical Chemistry A* 126.40 (2022), pp. 7407–7414. DOI: 10.1021/acs.jpca.2c05002. URL: https://doi.org/10.1021/acs.jpca.2c05002.

[9] Maximilian Puelma Touzel, Paul Cisek, and Guillaume Lajoie. "Performance-gated deliberation: A context-adapted strategy in which urgency is opportunity cost". In: *PLoS Computational Biology* 18.5 (May 2022), pp. 1–33. DOI: 10.1371/journal.pcbi.1010080. URL: https://doi.org/10.1371/journal.pcbi.1010080.

[8] Ryan Vogt, Maximilian Puelma Touzel, Eli Shlizerman, and Guillaume Lajoie. "On Lyapunov Exponents for RNNs: Understanding Information Propagation Using Dynamical Systems Tools". In: *Frontiers in Applied Mathematics and Statistics* 8 (2022). DOI: 10.3389/fams.2022.818799. URL: https://www.frontiersin.org/articles/10.3389/fams.2022.818799.

[7] Maximilian Puelma Touzel, Aleksandra M Walczak, and Thierry Mora. "Inferring the immune response from repertoire sequencing". In: *PLoS Computational Biology* 16.4 (2020), pp. 1–21. DOI: 10.1371/journal.pcbi.1007873. URL: http://dx.doi.org/10.1371/journal.pcbi.1007873.

[6] Maximilian Puelma Touzel and Fred Wolf. "Statistical mechanics of spike events underlying phase space partitioning and sequence codes in large-scale models of neural circuits". In: *Phys. Rev. E* 99 (May 2019), p. 52402. DOI: 10.1103/PhysRevE.99.052402. URL: https://link.aps.org/doi/10.1103/PhysRevE.99.052402.

[5] Susana Magadan, Luc Jouneau, Maximilian Puelma Touzel, Simon Marillet, Wahiba Chara, Adrien Six, Edwige Quillet, Thierry Mora, Aleksandra M Walczak, Frederic Cazals, Oriol Sunyer, Simon Fillatreau, and Pierre Boudinot. "Origin of Public Memory B Cell Clones in Fish After Antiviral Vaccination". In: *Frontiers in Immunology* 9 (2018). DOI: 10.3389/fimmu.2018.02115. URL: https://www.frontiersin.org/journals/immunology/articles/10.3389/fimmu.2018.02115.

[4] Mikhail V Pogorelyy, Anastasia A Minervina, Maximilian Puelma Touzel, Anastasiia L Sycheva, Ekaterina A Komech, Elena I Kovalenko, Galina G Karganova, Evgeniy S Egorov, Alexander Yu Komkov, Dmitriy M Chudakov, Ilgar Z Mamedov, Thierry Mora, Aleksandra M Walczak, and Yuri B Lebedev. "Precise tracking of vaccine-responding T cell clones reveals convergent and personalized response in identical twins". In: *Proc. Natl. Acad. Sci. U. S. A.* 115.50 (Dec. 2018), pp. 12704–12709. URL: https://www.pnas.org/doi/full/10.1073/pnas.1809642115.

[3] Carmen L Murall, Jessica L Abbate, Maximilian Puelma Touzel, Emma Allen-Vercoe, Samuel Alizon, Remy Froissart, and Kevin McCann. "Chapter Five - Invasions of Host-Associated Microbiome Networks". In: *Networks of Invasion: Empirical Evidence and Case Studies*. Ed. by David A Bohan, Alex J Dumbrell, and François Massol. Vol. 57. Advances in Ecological Research. Academic Press, 2017, pp. 201–281. DOI: 10.1016/bs.aecr.2016.11.002. URL: https://www.sciencedirect.com/science/article/pii/S0065250416300605.

[2] Maximilian Puelma Touzel and Fred Wolf. "Complete Firing-Rate Response of Neurons with Complex Intrinsic Dynamics". In: *PLoS Computational Biology* 11.12 (2015), pp. 1–43. DOI: 10.1371/journal.pcbi.1004636.

[1] Fred Wolf, Rainer Engelken, Maximilian Puelma Touzel, Juan Daniel Flórez Weidinger, and Andreas Neef. "Dynamical models of cortical circuits". In: *Current Opinion in Neurobiology* 25 (2014), pp. 228–236. DOI: 10.1016/j.conb.2014.01.017. URL: https://www.sciencedirect.com/science/article/pii/S0959438814000324.

Machine Learning, Multi-Agent & Complex Social Systems

[14] Mattia Greiche, Reihaneh Rabbany, and Maximilian Puelma Touzel. "A Framework for Context Enrichment and Selective Robustness in LLM Social Agents". In: *SocialAgent Workshop at NeurIPS*. 2026. URL: https://arxiv.org/abs/2611.06914.

[13] Maximilian Puelma Touzel, Sneheel Sarangi, Aurélien Bück-Kaeffer, Zachary Yang, Jean-François Godbout, and Reihaneh Rabbany. "Position: Time to Close The Validation Gap in LLM Social Simulations". In: *Forty-third International Conference on Machine Learning Position Paper Track*. 2026. URL: https://openreview.net/forum?id=LpbxLBcOBf.

[12] Matthew Riemer, Tommaso Tosato, Amin Memarian, Maximilian Puelma Touzel, Glen Berseth, Irina Rish, and Guillaume Dumas. "Position: Collusion Risks Among AI Reasoning Agents Justify Certification Requirements for Making Market Decisions". In: *Forty-third International Conference on Machine Learning Position Paper Track*. 2026. URL: https://openreview.net/forum?id=Vhi9tHbKTa.

[11] Zachary Yang, Anne Imouza, Maximilian Puelma Touzel, Cécile Amadoro, Gabrielle Desrosiers-Brisebois, Kellin Pelrine, Sacha Levy, Jean-François Godbout, and Reihaneh Rabbany. "Regional and temporal patterns of partisan polarization during the COVID-19 pandemic in the United States and Canada". In: *PLOS ONE* 21.4 (Apr. 2026), pp. 1–31. DOI: 10.1371/journal.pone.0347327. URL: https://doi.org/10.1371/journal.pone.0347327.

[10] Aurélien Bück-Kaeffer, Je Qin Chooi, Dan Zhao, Maximilian Puelma Touzel, Kellin Pelrine, Jean-François Godbout, Reihaneh Rabbany, and Zachary Yang. "BluePrint: A Social Media User Dataset for LLM Persona Evaluation and Training". In: *Workshop on Tailoring AI: Exploring Active and Passive LLM Personalization (PALS)*. EMNLP. 2025. URL: https://pals-nlp-workshop.github.io/.

[9] Taylor Lynn Curtis, Maximilian Puelma Touzel, William Garneau, Manon Gruaz, Mike Pinder, Li Wei Wang, Sukanya Krishna, Luda Cohen, Jean-François Godbout, Reihaneh Rabbany, and Kellin Pelrine. "Veracity: An Open-Source AI Fact-Checking System". In: *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*. 2025. DOI: 10.24963/ijcai.2025/1254.

[8] Maximilian Puelma Touzel, Sneheel Sarangi, Gayatri Krishnakumar, Busra Tugce Gurbuz, Austin Welch, Zachary Yang, Andreea Musulan, Hao Yu, Ethan Kosak-Hine, Tom Gibbs, Camille Thibault, Reihaneh Rabbany, Jean-François Godbout, Dan Zhao, and Kellin Pelrine. "SandboxSocial: A Sandbox for Social Media Using Multimodal AI Agents". In: *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*. 2025. DOI: 10.24963/ijcai.2025/1271.

[7] Hao Yu, Shenyang Huang, Zachary Yang, Maximilian Puelma Touzel, Kellin Pelrine, Jean-François Godbout, and Reihaneh Rabbany. "TRUTH: Teaching LLMs to Rerank for Truth in Misinformation Detection". In: *Third Workshop on Socially Responsible Language Modelling Research (SoLaR)*. COLM. 2025. URL: https://solar-colm.github.io/papers/.

[6] Maximilian Puelma Touzel and Erick Lachapelle. "Ideology from topic mixture statistics: inference method and example application to carbon tax public opinion". In: *Environmental Data Science* 3 (2024), e10. DOI: 10.1017/eds.2023.44.

[5] Maximilian Puelma Touzel, Amin Memarian, Matthew Riemer, Andrei Mircea, Andrew Robert Williams, Elin Ahlstrand, Lucas Lehnert, Rupali Bhati, Guillaume Dumas, and Irina Rish. "Scalable Approaches for a Theory of Many Minds". In: *ICML Agentic Markets Workshop*. 2024. URL: https://openreview.net/forum?id=P0oG5gDh6T.

[4] Maximilian Puelma Touzel, Sneheel Sarangi, Austin Welch, Gayatri Krishnakumar, Dan Zhao, Zachary Yang, Hao Yu, Ethan Kosak-Hine, Tom Gibbs, Andreea Musulan, Camille Thibault, Busra Tugce Gurbuz, Reihaneh Rabbany, Jean-François Godbout, and Kellin Pelrine. "A Simulation System Towards Solving Societal-Scale Manipulation". In: *NeurIPS Workshop on Socially Responsible Language Modelling Research*. 2024. URL: https://openreview.net/forum?id=fVl2Dhn4Kr.

[3] Maximilian Puelma Touzel, Amin Memarian, Matthew D Riemer, Rupali Bhati, and Irina Rish. "Summarizing Societies: Agent Abstraction in Multi-Agent Reinforcement Learning". In: *ICLR Workshop: From Cells to Societies: Collective Learning across Scales*. 2022. URL: https://openreview.net/forum?id=Sc9ESMyTZ9.

[2] Matthew Riemer, Sharath Chandra Raparthy, Ignacio Cases, Gopeshh Subbaraj, Maximilian Puelma Touzel, and Irina Rish. "Continual Learning In Environments With Polynomial Mixing Times". In: *Advances in Neural Information Processing Systems*. Ed. by S Koyejo, S Mohamed, A Agarwal, D Belgrave, K Cho, and A Oh. Vol. 35. Curran Associates, Inc., 2022, pp. 21961–21973. URL: https://proceedings.neurips.cc/paper_files/paper/2022/file/89c61fce5a8b73871d1c4073f486b134-Paper-Conference.pdf.

[1] Giancarlo Kerg, Kyle Goyette, Maximilian Puelma Touzel, Gauthier Gidel, Eugene Vorontsov, Yoshua Bengio, and Guillaume Lajoie. "Non-normal Recurrent Neural Network (nnRNN): learning long time dependencies while improving expressivity with transient dynamics". In: *Advances in Neural Information Processing Systems*. Ed. by H Wallach, H Larochelle, A Beygelzimer, F d'Alche-Buc, E Fox, and R Garnett. Vol. 32. Curran Associates, Inc., 2019. URL: https://proceedings.neurips.cc/paper_files/paper/2019/file/9d7099d87947faa8d07a272dd6954b80-Paper.pdf.

Quantum Information & Dissertation

[2] Maximilian Puelma Touzel. "Cellular dynamics and stable chaos in balanced networks". PhD thesis. University Goettingen, 2015. URL: http://dx.doi.org/10.53846/goediss-5477.

[1] Maximilian Puelma Touzel, Robert BA Adamson, and Aephraim M Steinberg. "Optimal bounded-error strategies for projective measurements in nonorthogonal-state discrimination". In: *Phys. Rev. A* 76 (Jan. 2007), p. 62314. DOI: 10.1103/PhysRevA.76.062314.

Teaching, Supervision & Outreach

Supervision

2025–present **Master's Co-supervisor**, *with Prof. Reihaneh Rabbany, McGill / Mila*, Montréal, Canada
2 Master's students

2021–present **PhD Co-supervisor**, *with Prof. Guillaume Lajoie, Université de Montréal / Mila*, Montréal, Canada
1 PhD student

2020–present **Master's Co-supervisor**, *with Prof. Irina Rish, Université de Montréal / Mila*, Montréal, Canada
1 Master's student

2018 **Master's Co-supervisor**, *with Profs. Aleksandra Walczak & Thierry Mora, ENS Paris*, Paris, France
1 Physics Master's student

2015 **Master's Co-supervisor**, *with Prof. Fred Wolf, MPI for Dynamics and Self-Organization*, Goettingen, Germany
1 Physics Master's student

2022–2024 **Research Manager**, *CERC in Autonomous AI, Mila*, Montréal, Canada
Day-to-day scientific mentoring of the group's graduate students and postdocs

Teaching

2020–2023 **Substitute Lecturer**, *MAT6215: Dynamical Systems (graduate), Université de Montréal*, Montréal, Canada

2022–2025 **Co-organizer & Content Creator**, *Climatematch Academy*, Virtual
Designed and delivered a day of lectures and computational tutorials on the socioeconomics of climate change for hundreds of students from more than 100 countries

2021 **Guest Lecturer**, *Physics and AI Reading Group, Mila*, Montréal, Canada
Stochastic thermodynamics of learning (26 May 2021; recording)

2019 **Workshop Tutor**, *Physics and AI Workshop*, Montréal, Canada

2014 **Tutorial Lecturer**, *Latin American Summer School in Computational Neuroscience*, Valparaíso, Chile
Theory and modelling methodology in biophysics, through case studies in computational neuroscience

2011–2015 **Teaching Assistant & Guest Lecturer**, *Graduate course in computational neuroscience (Prof. Fred Wolf), University of Goettingen*, Goettingen, Germany
Lectured on conductance-based neuron models, linear response theory of spiking neurons and phase-oscillator reductions

2013 **Workshop Instructor**, *Dynamical-systems training workshop for biologists*, Goettingen, Germany

2011–2015 **Course Co-coordinator & Content Manager**, *Seminar in Biophysics; Seminar in Theoretical Neuroscience*, University of Goettingen, Germany

2008–2009 **Tutorial Assistant**, *Department of Physics, University of Toronto*, Toronto, Canada
With Dr. David Harrison, head of the physics teaching faculty: redesigned tutorials around active-learning exercises informed by Physics Education Research; invited back for a second year

Outreach

2006–2008 **Science Educator & Communicator**, *Ontario Science Centre*, Toronto, Canada
Developed and performed public demonstrations on astronomy, robotics and resonance

2006 **Science Camp Coordinator**, *Activity science camp for Hispanic youth, Centre for Spanish-Speaking Peoples*, Toronto, Canada
Conceived, designed and ran a summer science camp for at-risk youth

2005 **Academic Tutor**, *Independent*, Toronto, Canada
Math, science and language support for newly immigrated youth

Service & Community Building

I build research communities across disciplines: I have founded or led four ongoing communities and schools, designed new formats for scientific exchange, and organized more than a dozen workshops, symposia and schools in physics, neuroscience and AI.

Reviewing *Physical Review X*, *Nature Human Behaviour*; *EMNLP*, *COLM*

Affiliations Member, Centre for the Study of Democratic Citizenship

Communities Founded & Led

2026–present **Founder & Coordinator**, *STAMINA working group (Social Tech And ModellIng for kNowledge and Action)*, Virtual
A working group for research standards, with a biweekly seminar series for fast communication of research results; complexdatalab.com/stamina

2022–2025 **Co-organizer**, *Climatematch Academy*, Virtual
International online summer school with students from more than 100 countries; led the socioeconomics curriculum in the 2024 edition and worked with the platform team on using AI to support the learning community at scale

2020–2022 **Co-organizer**, *Mila NeuroAI Reading Group*, Montréal, Canada
Scheduled and hosted international speakers; the group is still running

2012–2015 **Lead Organizer**, *Goettingen Advanced Course in Computational Neuroscience*, Goettingen, Germany
Led the organizing team; initiated, secured funding for and oversaw the transition to a new advanced-content format

New Formats for Scientific Exchange

2026 **Designer & Facilitator**, *"AI red penning" breakout session, LLM Social Simulation Workshop @ Conference on Language Modeling*, San Francisco, USA
Groups critique AI-generated research proposals (write-up); to be run again at AAAI 2027

2019 **Tutorial Co-designer**, *Hands-on machine learning tutorial, Montreal Physics and AI Workshop*, Montréal, Canada
Guided more than 100 participants, from high-school students to professors, through data-science exercises in shared notebooks

2019 **Group Discussion Designer**, *Mathematics of Vision Workshop, Fields Institute*, Toronto, Canada

2020 **Discussion Session Facilitator**, *UNIQUE Student Symposium*, Montréal, Canada
Higher-order cognition session

Workshops, Symposia & Schools Organized

2025 **Workshop Co-organizer**, *Social Simulation with LLMs @ Conference on Language Modeling*, Montréal, Canada

2023 **Session Chair & Representative to Conference Committee**, *AAAI Fall Symposium: Artificial Intelligence and Climate*, Washington, D.C., USA

2022 **Workshop Co-organizer**, *Social Alignment in Humans and Machines @ Reinforcement Learning & Decision-Making Conference*, Providence, USA

2021 **Symposium Co-organizer**, *Symposium on Explanation in Neuroscience & Artificial Intelligence*, Montréal, Canada

2019 **Workshop Co-organizer**, *NeurIPS Workshop: Real Neurons & Hidden Units*, Montréal, Canada
One-day event with >50 double-blind-reviewed papers, live video feed and panel

2019 **Conference Co-organizer (with N. Provatas)**, *Montreal Physics and AI Workshop, McGill University*, Montréal, Canada
200 participants; one of the first physics-and-AI meetings in Canada; lectures and beginner and advanced workshops

2017 **Symposium Co-organizer**, *Paris Biological Physics Community Day*, Paris, France

2014 **Summer School Co-coordinator**, *Latin American Summer School in Computational Neuroscience*, Valparaíso, Chile
Week 2: Network Neurodynamics, Instituto de Sistemas Complejos de Valparaíso

Research Talks

2026 **Invited Speaker**, *Surya Ganguli's group, Stanford University*, Stanford, USA
Role differentiation as ignition of a collective information engine

2026 **Speaker**, *Ross Otto Lab, McGill Psychology*, Montréal, Canada
Cognitive Effort & Identity in the Age of AI

2025 **Speaker**, *Mila AI Safety Reading Group*, Montréal, Canada
Schema-Resource Complexes for Evaluating Systemic Impact

2025 **Speaker**, *IJCAI*, Montréal, Canada
SandboxSocial: A Sandbox for Social Media Using Multimodal AI Agents

2025 **Speaker**, *Ross Otto Lab, McGill Psychology*, Virtual
Impact of psychological traits on effectiveness of AI-guided misinformation correction

2025 **Invited Speaker**, *Multi-Agent Seminar, Google DeepMind*, Virtual

2025 **Invited Speaker**, *Centre for the Study of Democratic Citizenship Seminar Series*, Montréal, Canada

2024 **Speaker**, *AAAI Workshop: AI & Climate: Role of AI in a Climate-Smart Sustainable Future*, Washington, D.C., USA

2023 **Invited Speaker**, *Lab Talk, Google DeepMind*, Virtual
Joel Leibo's research group

2022 **Invited Speaker**, *BIRS Workshop on Dynamical Principles of Biological & Artificial Neural Networks*, Banff, Canada

2021 **Speaker**, *Neural Scaling Laws Workshop*, Tremblant, Canada

2021 **Speaker**, *Reinforcement Learning Reading Group (Mila)*, Virtual

2020 **Speaker**, *Ross Otto Lab, McGill Psychology*, Virtual
Urgency as the opportunity cost of time

2020 **Speaker**, *NeuroAI Reading Group (Mila)*, Montréal, Canada
Inverse Rational Control

2019 **Invited Speaker**, *Soft Matter & Biophysics Seminar, Simon Fraser University*, Vancouver, Canada
An inference take on urgency in decision-making

2019 **Invited Speaker**, *Computational Neuroscience Seminar, University of Ottawa*, Ottawa, Canada
An inference take on urgency in decision-making

2019 **Invited Speaker**, *Quantitative & Computational Biology Seminar, Université de Montréal*, Montréal, Canada
Inferring repertoire dynamics from repertoire sequencing

2018 **Spotlight Speaker**, *q-bio Conference, Rice University*, Houston, USA
Ensemble response of immune repertoires to vaccination

2018 **Invited Speaker**, *Friday Seminar, UCL Gatsby Computational Neuroscience Unit*, London, UK
Understanding the shape of high-dimensional activity in cortex-inspired neural circuits

2018 **Invited Speaker**, *Biophysics Seminar, Emory University, Dept. of Physics*, Atlanta, USA

2018 **Speaker**, *APS March Meeting*, Los Angeles, USA
Repertoire-based approach to identifying sequence motifs specific to an effective vaccine

2018 **Invited Speaker**, ***Biophysics Seminar, McGill University, Dept. of Physics***, Montréal, Canada
Inferring contributions of recombination and selection to singly-perturbed repertoires

2018 **Invited Speaker**, *Tea Talk, Montreal Institute for Learning Algorithms*, Montréal, Canada
Don't paint the box black: Using dynamical systems to explain complex phase space geometry

2017 **Speaker**, *Systems Immunology and Vaccine Design Workshop*, Heidelberg, Germany
Repertoire-based approach to identifying sequence motifs specific to an effective vaccine

2017 **Speaker**, *Biophysics Seminar, University of Toronto, Dept. of Physics*, Toronto, Canada
Inferring contributions of recombination and selection to singly-perturbed repertoires

2016 **Speaker**, *PhD & PostDoc Seminar, ENS Dept. of Physics*, Paris, France
The statistical mechanics of phase space partitioning in large scale neuronal circuits

2015 **Speaker**, *Swartz Foundation Meeting*, Janelia Research Campus, USA
A theory for the balanced state that keeps track of each and every spike

2015 **Speaker**, *APS March Meeting*, San Antonio, USA
Elements of a finite-size ergodic theory for stable chaos

2015 **Speaker**, *ENS Theoretical Neuroscience Seminar*, Paris, France
A theory of precise spike timing in cortical circuits

2014 **Speaker**, *APS March Meeting*, Denver, USA
Microstate description of stable chaos in networks of spiking neurons

Selected Posters

2025 **Presenter**, *IJCAI*, Montréal, Canada

2022 **Presenter**, *NeurIPS Workshop on Tackling Climate Change with Machine Learning*, Virtual

2022 **Presenter**, *Montreal AI Symposium*, Montréal, Canada

2022 **Presenter**, *RLDM*, Providence, USA

2022 **Presenter**, *COSYNE*, Lisbon, Portugal

2021 **Presenter**, *NeurIPS EcoRL Workshop*, Virtual

2021 **Presenter**, *MAIS*, Virtual

2021 **Presenter**, *COSYNE*, Virtual
Urgency as the opportunity cost of commitment

2020 **Presenter**, *Biological and Artificial Reinforcement Learning Workshop, NeurIPS*, Virtual
Urgency as the opportunity cost of commitment

2020 **Presenter**, *Neuroscience and Artificial Intelligent Systems, Cold Spring Harbor Laboratory*, Virtual
Urgency as the opportunity cost of commitment

2020 **Presenter**, *COSYNE*, Denver, USA
An inference perspective on urgency in decision-making

2019 **Presenter**, *Montréal AI & Neuroscience Conference*, Montréal, Canada, Poster Prize Winner
An inference take on urgency in decision-making

2019 **Presenter**, *Physics & AI Workshop*, Montréal, Canada
Stochastic thermodynamics of aggregate-label learning

2018 **Presenter**, *Montréal AI & Neuroscience Conference*, Montréal, Canada
Transfer properties of multi-spike tempotrons

2018 **Presenter**, *q-bio Conference*, Houston, USA
Ensemble response of immune repertoires to vaccination

2018 **Presenter**, *Curie-Weizmann Meeting*, Paris, France
Inferring perturbations to immune repertoires using clone size statistics

2006–2017 16 further posters, including COSYNE (×2), Bernstein Conference (×3), Computational Neuroscience Society (×2; Poster Prize 2012), Les Houches *Dynamics and Information in Cells and Tissues*, Beg Rohu Summer School on Statistical Physics, Statistical Physics Methods in Biology and Computer Science (Paris), International Conference in Mathematical Neuroscience, and the International Conference on Quantum Information.

Advanced Schools & Programs

Biophysics KITP Santa Barbara programs on Quantitative Immunology (2016) and Olfaction (2015); Les Houches *Dynamics and Information in Cells and Tissues* (2016); Curie Institute *Multiscale Integration in Biological Systems* (2016); Beg Rohu Summer School on Statistical Physics (2017); Cargèse Summer School in Theoretical Biophysics (2017); IOP *Emergent Themes in Biophysics* (2007)

Neuroscience MBI *Mathematical Challenges in Neural Network Dynamics* (2013); Advanced Course in Computational Neuroscience (2012); Latin American Summer School in Computational Neuroscience (2009, 2014); CND Computational Neuroscience Summer School (2009); Complex Systems Summer School, Valparaíso (2008)

AI IVADO *Autonomous LLM Agents* bootcamp (2025); Mila TRAIL course in AI ethics (2023)

Methods & Tools

Theory Statistical mechanics, stochastic processes, Bayesian inference, dynamical systems, reinforcement learning

Computation Python ML & data-science stack (PyTorch, scikit-learn, pandas, NumPy, Weights & Biases); cluster computing (Slurm); LaTeX/TikZ

Languages

English (native), Spanish (fluent), French (conversational)